Quiz Bowl RL Buzzer

Learning When to Buzz: Reinforcement Learning for MC Quiz Bowl Stopping

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github.com/hass0114/qanta-buzzer

Problem

- Quiz bowl questions reveal clues incrementally — early clues are obscure, later clues are easy (pyramidal structure).
- The challenge is when to buzz, not just what to answer. Buzz too early □ wrong; too late □ wasted opportunity.
- Standard QA benchmarks measure only final accuracy, ignoring uncertainty and confidence calibration.
- We study this in a multiple-choice setting so the answer space is controlled and evaluation is reproducible.

Background

- Modeled as a POMDP: at each timestep the agent observes a partial clue prefix and chooses WAIT or BUZZ(answer).
- K=4 options: one correct answer + three artifact-resistant distractors.
- Two policy families: belief-feature buzzers (threshold, softmax-profile, Bayesian, PPO) and end-to-end T5 text-policy buzzers.

Why Multiple Choice?

- Open-ended QA adds aliasing and grading complexity — MC isolates the buzzing decision itself.
- Naive distractors create artifacts: models exploit surface-level answer quirks rather than using clues.
- Anti-artifact guards: alias collision checks, token overlap, length ratio, question-text overlap screening.

Our Contribution

- First application of PPO to MC pyramidal quiz bowl (novel POMDP + RL combination).
- Artifact-resistant MC construction with multiple distractor strategies.
- Calibration-focused evaluation: S_q, ECE, Brier, plus choices-only / shuffle / alias controls.

POMDP Formulation

State: full question + answer key (hidden)
Observation: clue prefix h_t + MC option set C
Actions: WAIT or BUZZ(i) for i □ {A,B,C,D}
Reward: +1 correct buzz, −1 wrong, 0 wait (shaped)

Method

1. Load tossup questions from QANTA dataset
2. Build answer profiles from historical text
3. Construct artifact-resistant MC questions
4. Score options with likelihood models (TF-IDF / SBERT / T5)
5. Convert beliefs into structured observations
6. Run baseline or learned RL buzzers
7. Evaluate: timing, accuracy, calibration, controls

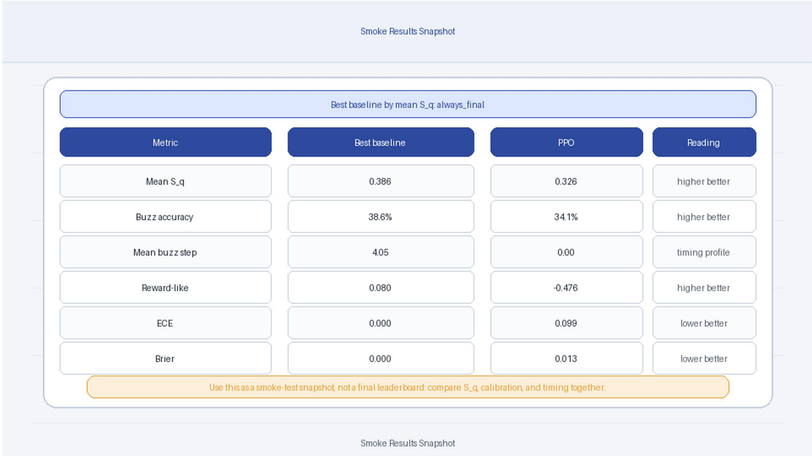
Factored Policy

- Answer module □ p_ans(i | h_t): which answer is best?
- Stop module □ p_buzz(h_t): should I buzz now?
- Composition: P(WAIT) = 1−p_buzz; P(BUZZ i) = p_buzz × p_ans(i)

Training Recipe

Step 1: Supervised answer model — learn p_ans(i) via cross-entropy
Step 2: Hazard-loss warm start — learn when to commit
Step 3: PPO fine-tuning — optimize task reward (correct + timing bonus)

Posterior Sharpening



Posterior at buzz: A reaches 88% □ agent commits

Optimal stopping rule:
BUZZ when V_commit(h_t) > V_wait(h_t)
Commit value rises as posterior sharpens; wait value falls as remaining information decreases.

Experiments

- Task: given partial clue prefix + 4 MC options, decide WAIT or BUZZ(answer).
- Input: structured belief vector (posterior, max prob, entropy, step, prefix-option similarity scores).
- Output: action □ {WAIT, BUZZ_A, BUZZ_B, BUZZ_C, BUZZ_D}.
- Baselines: threshold, softmax-profile, sequential Bayes, always-final.

Results

Metric	Supervised	PPO (50 iter)	Direction
Accuracy	68%	64%	□ (intended)
Avg Reward	+0.31	+0.42	□ +35%
Buzz Position	2.3 clues	3.1 clues	□ cautious
ECE	0.18	0.15	□ better
Early Acc	41%	—	
Late Acc	79%	—	

Key insight:
PPO trades raw accuracy for better timing and calibration — lower accuracy + better timing = higher reward.

Analysis

- Commit value rises as posterior sharpens; wait value falls as remaining information decreases.
- Optimal buzz at the crossing point — RL agent learns this threshold.
- Controls: choices-only baseline (9.1% acc, below 25% chance) confirms model uses clues, not answer artifacts.
- Shuffle test: reordering clues does not degrade performance, confirming the agent reads content, not position.
- Alias substitution: replacing gold-answer text with synonyms preserves accuracy, ruling out string-matching shortcuts.

Abstain ≠ NOTA:
When the agent abstains it means 'wait for one more clue,' not 'none of the above.' The gold answer is always present.

Conclusions

- PPO learns meaningful buzzing policies that improve reward and calibration over supervised baselines.
- Factored policy (answer + stop module) simplifies RL and isolates the timing decision.
- Artifact controls validate that improvements come from clue understanding, not surface patterns.

References

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[3] Balepur et al. Artifacts or Abduction: LLMs Cheat on MC Benchmarks. ACL 2024.